

ABHILASH NEOG

🌐 [abhilash-neog.github.io](https://github.com/abhilash-neog) [in](#) LinkedIn [🎓](#) Scholar [G](#) Github [✉](mailto:abhilash22@vt.edu) abhilash22@vt.edu

Research Summary

Ph.D. researcher in machine learning with experience developing foundation models, generative models, and production ML systems. Work spans model architecture and multi-GPU training to ML pipelines and deployment, with recent research in diffusion and flow models, multimodal learning, and time-series foundation models.

Education

Virginia Tech

Ph.D. in Computer Science

M.S. in Computer Science

Advisor: Dr. Anuj Karpatne | GPA: 4.0/4.0

Blacksburg, VA

Aug 2022 – May 2027

Aug 2022 – Dec 2024

Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering (B.E.) in Computer Science, GPA: 8.08/10

Pilani, India

July 2016 – July 2020

Work Experience

Applied Scientist Intern | [Microsoft \(Ads & Monetization\)](#)

May 2026 – Aug 2026

- Designed and trained a FLUX-based generative AI system using LoRA fine-tuning and adaptive supervision, improving FID by 7.7% over state-of-the-art baselines.
- Built an automated 12K-image data curation and evaluation pipeline integrating GPT-5.5, prompt-based segmentation models, and quality filtering.
- Built a MLLM-as-a-Judge framework for automated evaluation of image generations across multiple outpainting rubrics, with method anonymization and randomized candidate ordering to reduce evaluator leakage and position bias.
- Built an end-to-end generative AI pipeline integrating VLM-based semantic guidance, segmentation, feature estimation, and image conditioning while preserving base-model inference cost and latency

Machine Learning Intern | [Kryptowire](#)

May 2023 – Aug 2023

- Developed an outlier-detection model to denoise sensor-based Human Activity Recognition (HAR) time-series data
- Built and deployed a Keras CNN for HAR to an Android application using TensorFlow Lite, achieving an 82% F1 score

Data Scientist | [Oracle](#)

Sep 2020 – July 2022

- Built and deployed production ML applications for demand forecasting, AR prediction, and AR delay prediction, spanning data processing, model training, evaluation, and production integration within Spark, MLOps, and CI/CD pipelines.
- Developed a large-scale classification system over a 71K-label taxonomy, improving accuracy 40% over prior LLM-based approaches

Software Engineer Intern | [VMware](#)

Jan 2020 – June 2020

- Streamlined Workspace ONE Cloud data workflows using Spring Boot REST APIs, and contributed to an end-user federation application with JUnit/Mockito testing

Software Engineer Intern | [Samsung Research Institute](#)

May 2019 – July 2019

- Optimized the C++ HAL call flow for Live Focus under low-light conditions, achieving considerable noise reduction during the first phase of camera capture

Research Experience

Graduate Research Assistant | [Knowledge-guided Machine Learning Lab](#)

Jan 2023 – Present

- Developing LLM-conditioned foundation-model systems that combine language context with time-series representations using diffusion transformers and flow matching for controllable generation.
- Designed and trained LakeFM, a foundation model for irregular multivariate time series, with custom tokenization and query-based forecasting, and multi-GPU training, achieving 22% lower average MSE error than the strongest competing foundation models
- Contributed to a physics-guided Diffusion Model for solving Partial Differential Eqns., achieving $15\text{-}250\times$ faster inference than SOTA baselines
- Vision-Language Model (VLM) evaluation for visual reasoning, fine-grained attribute grounding and zero-shot scene graph generation; contributed to fine-grained segmentation dataset generation and segmentation model benchmarking
- Designed and developed Model-agnostic Adapters for Irregular and Sparse Time-Series modeling

Selected Publications (*denotes equal contribution)

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen, Jian Jiao. “Preserving Subject-Clarity in Image Outpainting with Multiscale Wavelet Supervision”. [Under Review](#)
2. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. “LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data”. [KDD 2026](#)
3. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. “Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling”. [TMLR 2026](#)
4. A. Pandey*, **Abhilash Neog***, Gautam Jajoo. “On the Internal Semantics of Time-Series Foundation Models”. [BERT2S @ NeurIPS 2025](#)
5. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. “PRISMA: Improving the Accuracy–Latency Frontier of Diffusion-based PDE Solvers Using Physics-Informed Spectral Attention”. [Under Review](#)
6. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. [\(Oral\) CV4Science @ CVPR 2025](#)
7. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. “Open World Scene Graph Generation using Vision Language Models”. [CVinWild @ CVPR 2025](#)
8. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images”. [CVPR 2025](#)
9. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images”. [NeurIPS 2024](#)

Patents

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen. [Microsoft, 2026](#). “Subject-Clarity-Aware Image Outpainting via Semantic and Structural Guidance with Vision-Language Models and High-Frequency Map Estimation.” [Patent submission \(Under Review\)](#).
2. Akash B., Krishnan R., **Abhilash, N.**, Dipawesh P. and Karthik B. [Oracle, 2024](#). “Machine Learning Based Spend Classification.” [U.S. Patent Application 17/903,161](#).

Talks/Awards

- Invited talk at the [ASLO 2026 Conference - Session on AI for the Next Frontier in Aquatic Sciences](#)
- Invited talk at the [AGU 2025 Conference - Session on Advancing Water Science Through AI](#)
- [NSF NAIRR \(National AI Research Resource\) Pilot Award, 2024](#)
- “*Star of the Month (Dec 2021)*” within the Oracle Analytics Cloud Organization, Oracle India

Academic Service

- **Reviewing:**
 - Conferences/Journal - NeurIPS ('25, '26), AAAI ('26, '27), TMLR, ICML 2026, KDD 2026, ECCV 2026
 - Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)

Technical Skills

AI/ML: PyTorch, HuggingFace Transformers, Diffusers, vLLM, LoRA/PEFT, foundation models, LLMs/VLMs, diffusion models
ML Systems: multi-GPU training, model evaluation, data pipelines, inference pipelines, Spark, MLOps, CI/CD
Languages/Software: Python, C++, SQL, Git, REST APIs, Hydra